Since $\triangle POR$ and $\triangle SOQ$ are congruent. So, $PR = SQ$ which gives $PR^2 = SQ^2$

Thus, $[\cos(\alpha+\beta) - 1]^2 + \sin^2(\alpha+\beta) = [\cos\alpha - \cos(-\beta)]^2 + [\sin\alpha - \sin(-\beta)]^2$

$$-2\cos(\alpha+\beta) + 2 = 2 - 2\cos\alpha\cos\beta + 2\sin\alpha\sin\beta$$

Hence, $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$.

(i) In the above proof, $PR = SQ$ says that the distance between two points on a circle is determined by the radius and the central angle .

(ii) Arc lengths PR and SQ, subtends angles $\alpha+\beta$ and $\alpha+(-\beta)$ respectively at the center. Thus, PR = SQ. Thus, distance between the points $(\cos\alpha, \sin\alpha)$ and $(\cos(-\beta), \sin(-\beta))$ is same as the distance between the points $(\cos(\alpha+\beta), \sin(\alpha+\beta))$ and $(1, 0)$.

(iii) In the above derivations, $0 \le \alpha < 2\pi,\ 0 \le \beta < 2\pi$. Because of periodicity of sine and cosine, the result follows for any α and β.

Identity 3.2: $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

Proof.

We know that $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$$\begin{aligned}\text{Now,}\quad \cos(\alpha-\beta) &= \cos[\alpha+(-\beta)]\\ &= \cos\alpha\cos(-\beta) - \sin\alpha\sin(-\beta)\\ \text{Hence,}\ \cos(\alpha-\beta) &= \cos\alpha\cos\beta + \sin\alpha\sin\beta.\end{aligned}$$

(i) If $\alpha = \beta$, the above identity is reduced to $\cos^2\alpha + \sin^2\alpha = 1$.

(ii) If $\alpha = 0$ and $\beta = x$, then $\cos(-x) = \cos x$, which shows that $\cos x$ is an even function.

Identity 3.3: $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Proof. This formula may be proved by writing $\sin(\alpha+\beta) = \cos\left[\frac{\pi}{2} - (\alpha+\beta)\right] = \cos\left[\left(\frac{\pi}{2} - \alpha\right) - \beta\right]$ and by using Identity 3.2.

If $\alpha+\beta = \frac{\pi}{2}$, the above identity is reduced to $\cos^2\alpha + \sin^2\alpha = 1$.

Identity 3.4: $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

Proof. This formula may be proved by writing $\sin(\alpha-\beta) = \sin[\alpha+(-\beta)]$ and by using Identity 3.3.

The sum and difference formulas for sine and cosine can be written in the matrix form

$$\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}\cdot\begin{pmatrix}\cos\beta & -\sin\beta\\ \sin\beta & \cos\beta\end{pmatrix} = \begin{pmatrix}\cos(\alpha+\beta) & -\sin(\alpha+\beta)\\ \sin(\alpha+\beta) & \cos(\alpha+\beta)\end{pmatrix}.$$

Identity 3.5: $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$

Proof.

$$\tan(\alpha+\beta) = \frac{\sin(\alpha+\beta)}{\cos(\alpha+\beta)} = \frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta - \sin\alpha\sin\beta}$$